

The Full FTP Rate — Why We Built It This Way

`docs/base_curve_construction.md` and `docs/liquidity_premium_construction.md` walk through *how* we build $\text{Base Rate} + \text{LP} = \text{Full FTP Curve}$. This is the companion piece — *why* we built it as two separately-fitted curves instead of one, and why each individual modelling choice inside that sits where it does, across the maths, the financial logic, the software design, and the business strategy it actually serves.

Decision: two separately-fitted curves, not one combined fit

Mathematically — the base curve and the liquidity premium are answering two different questions. The base curve asks “what does pure term risk cost, with no liquidity consideration at all” (answered from traded, liquid instruments — T-bills and bonds). The liquidity premium asks “what extra spread does a less-liquid funding source demand over that.” Fitting both in one pass would blur two economically distinct signals into one curve shape, and make it much harder to diagnose which one is actually driving a pricing outcome.

Financially — this decomposition is what lets us manage rate risk and liquidity risk as the separate risk categories they actually are. Term/duration risk gets hedged and managed differently from liquidity risk; a combined curve would obscure how much of any given price is compensation for one versus the other. This is precisely the failure mode an international survey of large-bank liquidity transfer pricing practice flagged as widespread: banks that priced liquidity at one blended average rate, rather than attributing it separately from term cost, ended up failing to properly penalise long-term funding commitments or reward long-term funding benefits — and, in aggregate, encouraged more maturity transformation than the bank’s own risk appetite would otherwise support (Grant, 2011).

In the software — `fit_base_curve` and `fit_liquidity_premium` are two independent methods on `CurveModelObject`, each with its own bounds, weights, and starting guess, callable and testable on their own. That separation is also what makes the liquidity premium’s own stress overlay possible without touching the base curve at all.

As a business decision — this is a genuinely useful management lens. Being able to see how much of our margin comes from term positioning versus how much comes from being a reliable, liquid funding source lets us make deliberate strategic calls — for instance, whether to compete harder for “sticky” deposits (a liquidity-premium play) or to lean into duration matching (a base-curve play) — rather than only seeing one blended number and guessing

at what's behind it.

Decision: Nelson-Siegel as the curve-fitting technique

Mathematically — Nelson-Siegel is a parsimonious, four-parameter model (level, slope, curvature, decay) that reliably produces smooth, economically sensible curve shapes from a limited number of observed points, without overfitting noise in the underlying data — the original formulation explicitly targeted exactly the monotonic, humped, and S-shaped curves observed in practice, and demonstrated the model could explain the large majority of yield variation across maturities from a small, stable parameter set (Nelson & Siegel, 1987).

Financially — a smooth, well-behaved curve matters beyond aesthetics: an overfitted curve with unstable kinks can create internal pricing anomalies — points where one tenor is priced inconsistently relative to its neighbours — that a business unit could exploit by structuring around the kink rather than genuine economics.

In the software — we reuse the same core fitting machinery (`fit_ns` , `ns_curve` , `ns_yield`) for both the base curve and the liquidity premium, just with different bounds, weights, and constraints passed in. That's deliberate reuse rather than two parallel implementations to maintain.

As a business decision — Nelson-Siegel is an established, externally-recognized technique, not something built from scratch. That matters when the curve needs to be explained to ALCO, audited, or benchmarked against how peer institutions build the same thing — it's defensible on its own terms, not a black box unique to us.

Decision: the 3-month shift as optional, off by default

Mathematically — this is an anchor-point correction: taking the fitted curve and nudging it to match one specific, highly liquid observed point (the 3-month deposit-vs-T-bill spread) rather than trusting the unconstrained fit at that tenor. That's a genuinely different modelling choice from the base fit itself, which is why we keep it as a distinct, separable step rather than baking it into the fit.

Financially — enabling it ties our short-end pricing more tightly to what we could actually achieve in the market today; disabling it lets the broader curve fit stand on its own, less reactive to short-term noise in one specific deposit comparison.

In the software — a config toggle (`CurveConfig`), off by default, rather than a code branch someone has to remember to guard. Turning it on or off is a one-line, reversible decision made explicitly per run, not an implicit side effect of which function got called.

As a business decision — this is genuinely a pricing-philosophy choice, not just a technical

knob: do we want our internal funding curve to react to short-end competitive deposit pricing, or do we want it insulated from that noise and driven purely by the broader market curve? Different institutions, and different points in a rate cycle, can reasonably land on different answers — which is exactly why we built it as something to decide deliberately, not something hardcoded.

Decision: a capped, separable stress overlay on the liquidity premium

Mathematically — the stress overlay is a multiplicative function of tenor, added as its own component on top of the structural LP rather than blended into it, and explicitly capped in basis points so a single extreme input can't dominate the fit.

Financially — this prices the cost of funding under stressed conditions, not just current conditions — the liquidity premium curve becomes a tool for scenario testing, not only a point-in-time snapshot.

In the software — `apply_stress` is a boolean, and the stress contribution is tracked as its own diagnostic column (`Stress_LP`) alongside the structural component, so a reviewer can see exactly how much of any given LP figure is "structural, observed today" versus "stress adjustment on top."

As a business decision — this connects retail and wholesale pricing directly to Treasury's own liquidity risk appetite. A bank that wants every loan and deposit price to already reflect a stressed view of funding cost — not just today's — has a direct, auditable mechanism to do that, rather than managing liquidity stress as a separate exercise disconnected from day-to-day pricing, in the same spirit as the post-crisis liquidity risk monitoring framework that pushed banks toward holding and pricing for genuinely stressed funding conditions rather than business-as-usual ones (Basel Committee on Banking Supervision [BCBS238], 2013).

See also `docs/base_curve_construction.md` and `docs/liquidity_premium_construction.md`.

References

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